



Advanced Web Development

Members:

- Simbaña Michael
- Tufiño Erick
- Villagómez Doménica

NRC: 27819

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How to obtain and use an authentication token

Use:

POST

medical-appointment-backend-2xx0.onrender.com/api/auth/login

Request

```
{  
  "email": "paciente@clinica.com",  
  "password": "admin123"  
}
```

The patient, doctor, or administrator account will be chosen according to the URI (The JSON remains the same, only "paciente" changes to "doctor" or "admin". The password is the same)

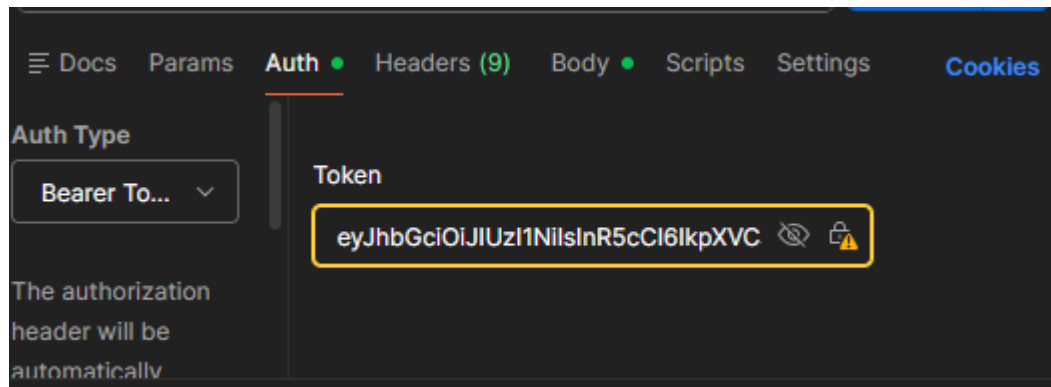
Response

```
{  
  "token":  
  "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VySWQiOiJjNDdlODQ0Yy0xNTYzLTQzMTQtOWFhYS00NGQ2YzUxOGEyYjciLCJyb2xlIjoicGF0aWVudCIsImVtYWlsIjoicGFjaWVudGVAY2xpbmljYS5jb20iLCJpYXQiOiJlbnV4MTM5OTcsImV4cCI6MTc2NjIwMDM5N30.jQghAw8OL2hcZY53je43WiAWMa6kpA1MGg7aRHDQpDw",  
  "user": {  
    "id": "c47e844c-1563-4314-9aaa-44d6c518a1b7",  
    "email": "paciente@clinica.com",  
    "role": "patient",  
    "first_name": "Carlos",  
    "last_name": "Mendoza",  
    "phone_number": null,  
    "cedula": null  
  }  
}
```

Copy the token obtained

```
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VySWQiOiJjNDdlODQ0Yy0xNTYzLTQzMTQtOWFhYS00NGQ2YzUxOGEyYjciLCJybn2xIjoicGF0aWVudCI6ImVtYWlsIjoicGFjaWVudGVAY2xpbnljYS5jb20iLCJpYXQiOiJlE3NjYxMTM5OTcsImV4cCI6MTc2NjIwMDM5N30.jQghAw8OL2hcZY53je43WiAWMa6kpA1MGg7aRHDQpDw
```

And paste it into the Auth tab in Postman. Select Bearer Token and put it in the token textbox.



After that, you will be able to use the methods belonging to that role normally.

Feature 1: Appointment Management and Notifications

Req. ID	Grouped Requirement Description	Roles Involved	Type
RF-P.3, RF-P.4	Appointment Search by Doctor/Specialty/Date and Dynamic Available Slot Calculation.	Patient	Functional (Query/Calculation)
RF-P.5	Booking a New Appointment by the Patient.	Patient	Functional (Create Appointment)
RF-P.6	Modification or Cancellation of an Existing Appointment by the Patient.	Patient	Functional (Update/Delete Appointment)
RF-D.1, RF-D.2	Work Schedule Configuration and Display of the Doctor's Consolidated Calendar.	Doctor	Functional (Schedule CRUD/View)

Feature 1:

Book a new appointment for the patient (Paciente role)

Post /api/appointments

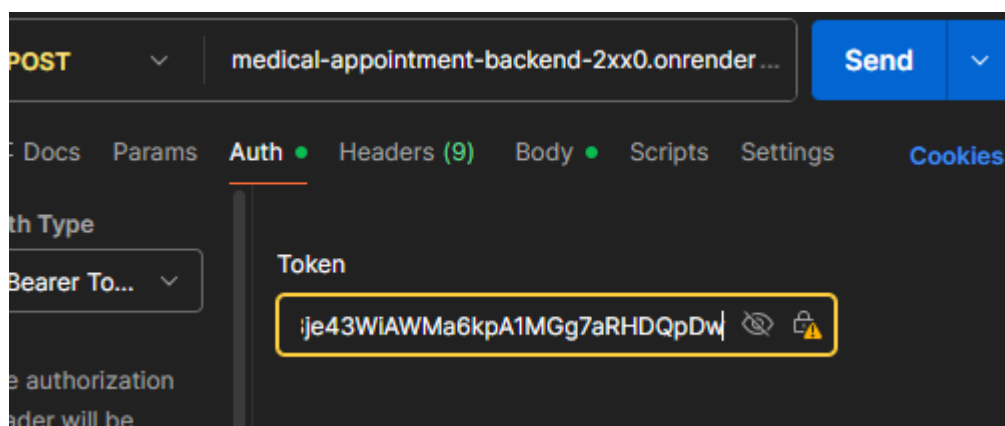
medical-appointment-backend-2xx0.onrender.com/api/appointments

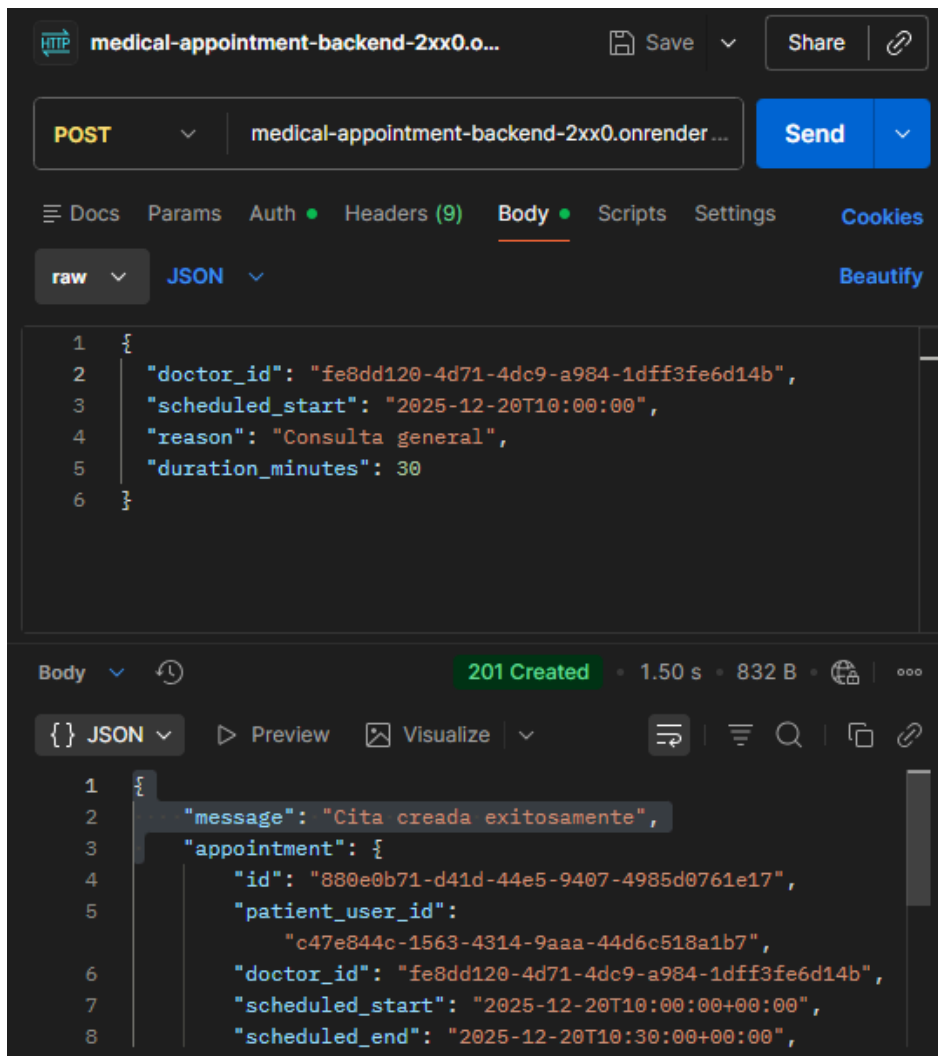
Request

```
{
  "doctor_id": "fe8dd120-4d71-4dc9-a984-1dff3fe6d14b",
  "scheduled_start": "2025-12-29T10:00:00",
  "reason": "Consulta general",
  "duration_minutes": 30
}
```

Response

```
{
  "message": "Cita creada exitosamente",
  "appointment": {
    "id": "4f664bf9-7a3c-416a-88e6-777b4efb05dc",
    "patient_user_id": "c47e844c-1563-4314-9aaa-44d6c518a1b7",
    "doctor_id": "fe8dd120-4d71-4dc9-a984-1dff3fe6d14b",
    "scheduled_start": "2025-12-29T10:00:00+00:00",
    "scheduled_end": "2025-12-29T10:30:00+00:00",
    "status_id": 1,
    "reason": "Consulta general",
    "created_at": "2025-12-19T05:00:53.422053+00:00",
    "updated_at": "2025-12-19T05:00:53.422053+00:00",
    "room_id": "b1a7f0a0-9c1a-4f8e-a9d1-3e4c6b8a2e1f",
    "created_by_user_id": "c47e844c-1563-4314-9aaa-44d6c518a1b7"
  }
}
```





Appointment Search by Doctor/Specialty/Date and Dynamic Calculation of Available Slots (Patient role)

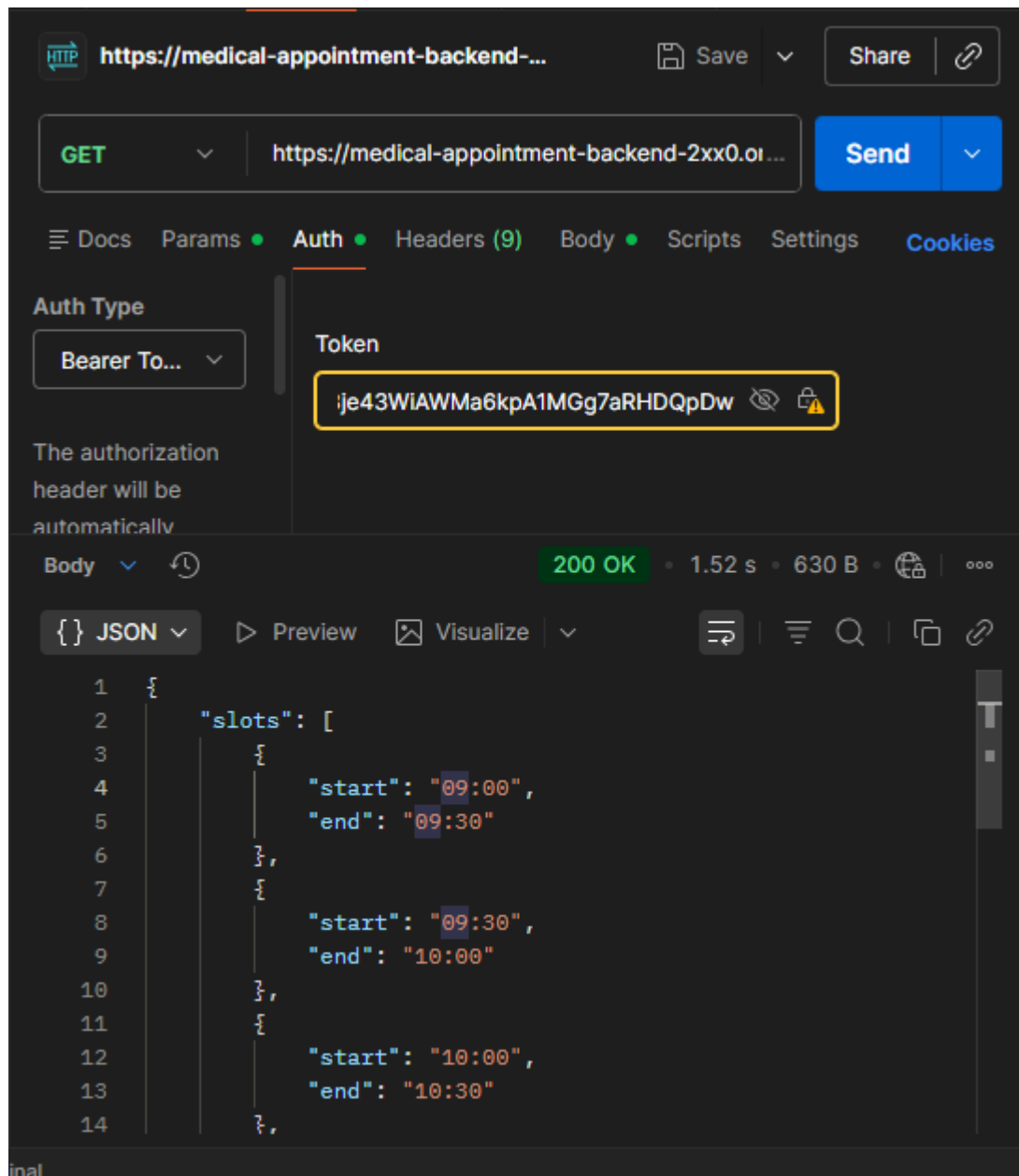
GET /api/appointments/doctors/:doctorId/available-slots?date=YYYY-MM-DD

medical-appointment-backend-2xx0.onrender.com/api/appointments/doctors/fe8dd120-4d71-4dc9-a984-1dff3fe6d14b/available-slots?date=2025-12-23

Response

```
{
  "slots": [
    {
      "start": "09:00",
      "end": "09:30"
    },
    {
      "start": "09:30",
      "end": "10:00"
    },
  ]
}
```

```
        "start": "10:00",
        "end": "10:30"
    },
    {
        "start": "10:30",
        "end": "11:00"
    },
    {
        "start": "11:00",
        "end": "11:30"
    },
    {
        "start": "12:00",
        "end": "12:30"
    },
    {
        "start": "12:30",
        "end": "13:00"
    },
    {
        "start": "13:00",
        "end": "13:30"
    },
    {
        "start": "13:30",
        "end": "14:00"
    }
]
}
```



Appointment Modification or Cancellation by the Patient (Patient role)

PUT `/api/appointments/:id`

medical-appointment-backend-2xx0.onrender.com/api/appointments/123e4567-e89b-12d3-a456-426614174000

Request

```
{
  "reason": "Control de presión arterial y revisión de medicamentos"
}
```

Response

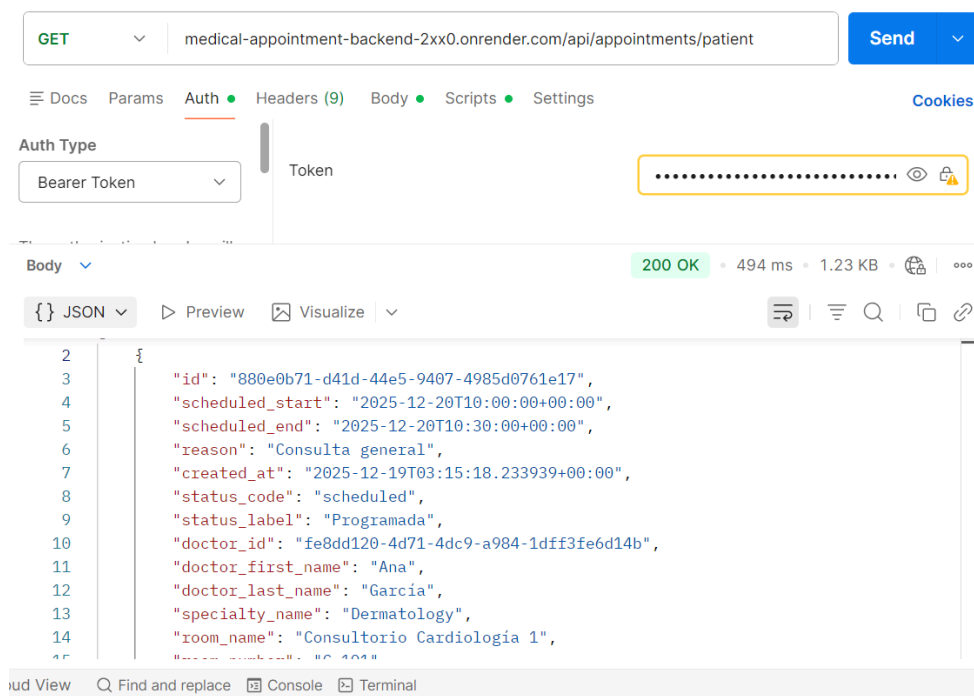
```
{
```

```

"message": "Cita creada exitosamente",
"appointment": {
  "id": "4f664bf9-7a3c-416a-88e6-777b4efb05dc",
  "patient_user_id": "c47e844c-1563-4314-9aaa-44d6c518a1b7",
  "doctor_id": "fe8dd120-4d71-4dc9-a984-1dff3fe6d14b",
  "scheduled_start": "2025-12-29T10:00:00+00:00",
  "scheduled_end": "2025-12-29T10:30:00+00:00",
  "status_id": 1,
  "reason": "Consulta general",
  "created_at": "2025-12-19T05:00:53.422053+00:00",
  "updated_at": "2025-12-19T05:00:53.422053+00:00",
  "room_id": "b1a7f0a0-9c1a-4f8e-a9d1-3e4c6b8a2e1f",
  "created_by_user_id": "c47e844c-1563-4314-9aaa-44d6c518a1b7"
}
}

```

Soporte Paciente — GET /api/appointments/patient (patient)



Soporte Paciente — GET /api/appointments/patient/:id (patient)

-?

Soporte Doctor — GET /api/appointments/doctor/:id (doctor)

Uri: medical-appointment-backend-2xx0.onrender.com/api/appointments/doctor

Request:

```
{
  "id": "fe8dd120-4d71-4dc9-a984-1dff3fe6d14b"
}
```

The screenshot shows a REST client interface with a GET request to `medical-appointment-backend-2xx0.onrender.com/api/appointments/doctor`. The response is a JSON object with an `id` field. Below the request, the 'Body' tab is selected, showing a 200 OK status with a response time of 869 ms and a body size of 832 B. The response body is a JSON array containing one object with various appointment details.

```
GET medical-appointment-backend-2xx0.onrender.com/api/appointments/doctor
```

raw JSON

```
1 {
2   "id": "fe8dd120-4d71-4dc9-a984-1dff3fe6d14b"
3 }
4
```

Body 200 OK • 869 ms • 832 B

```
1 [
2   {
3     "id": "761b9ac6-5932-4618-b258-4563ca6de09e",
4     "patient_name": "Jhon Doe",
5     "patient_email": "prueba@clinica.com",
6     "scheduled_start": "2025-12-22T15:30:00+00:00",
7     "scheduled_end": "2025-12-22T16:00:00+00:00",
8     "reason": "Seguimiento",
9     "status_id": 1,
10    "status_label": "Programada",
11  }
12 ]
```

Soporte Doctor — POST `/api/appointments/doctor/create` (doctor)

Requires Token

Uri: medical-appointment-backend-2xx0.onrender.com/api/appointments/doctor/create

Request:

```
{
  "patient_user_id": "c148742c-e93e-44d5-9660-161c6d81a54f",
  "scheduled_start": "2025-12-25T15:30:00+00:00",
}
```

POST

medical-appointment-backend-2xx0.onrender.com/api/appointments/doctor/create

Send

Docs

Params

Auth

Headers (9)

Body

Scripts

Settings

Cookies

raw

JSON

Beautify

```

1  {
2      "patient_user_id": "c148742c-e93e-44d5-9660-161c6d81a54f",
3      "scheduled_start": "2025-12-30T15:30:00+00:00"
4  }
5

```

Body

201 Created • 1.27 s • 829 B •

JSON

Preview

Visualize

```

1  {
2      "message": "Cita agendada exitosamente",
3      "appointment": {
4          "id": "daae17b3-2fcc-40b8-9134-3eb59f1e3899",
5          "patient_user_id": "c148742c-e93e-44d5-9660-161c6d81a54f",
6          "doctor_id": "e3605d0f-5a47-4d97-b43a-863f7c8090e7",
7          "scheduled_start": "2025-12-30T15:30:00+00:00",
8          "scheduled_end": "2025-12-30T16:00:00+00:00",
9          "status_id": 1,
10         "reason": null,

```

Find and replace

Console

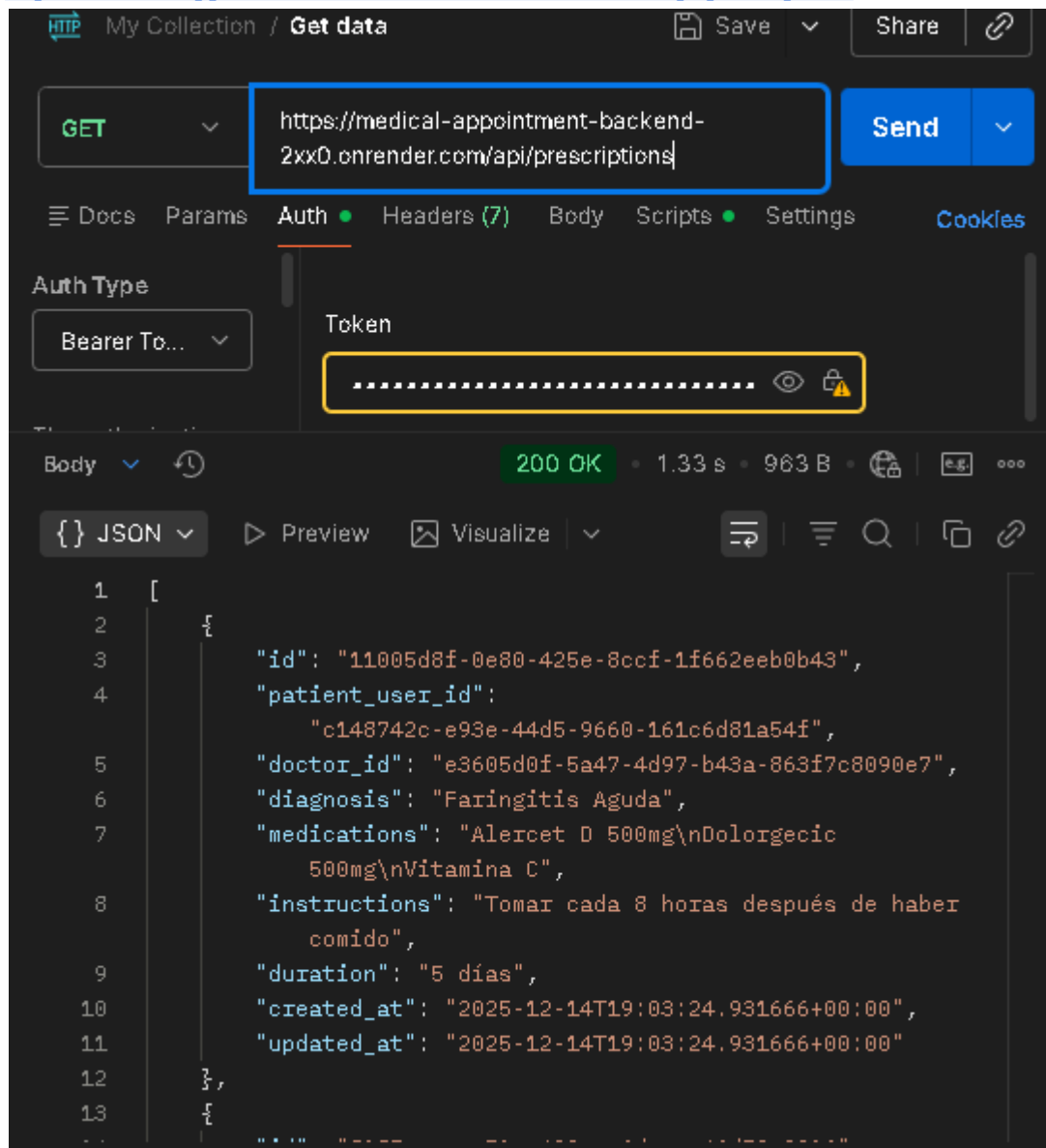
Terminal

Feature 2: Consultation and Prescription Management

Req. ID	Grouped Requirement Description	Roles Involved	Type
RF-D.3	Secure Access to the Patient’s Electronic Health Record (EHR) before consultation.	Doctor	Functional (Integration/Read)
RF-D.4	Update of Appointment Status (In Progress, Completed, etc.).	Doctor	Functional (Status Update)
RF-D.6	Creation and Digital Signing of the Medical Prescription (Receta Médica).	Doctor	Functional (Prescription/Create)
RF-D.5	Attachment of Diagnosis/Prescription Notes to the EHR after consultation.	Doctor	Functional (Integration/Update)

Feature 2:

<https://medical-appointment-backend-2xx0.onrender.com/api/prescriptions>



Soporte Doctor — PATCH /api/appointments/:id/status (doctor)

Requires Token

Uri:

medical-appointment-backend-2xx0.onrender.com/api/appointments/182e2dc0-d877-41cc-8676-7203074f92fc/status

Request:

```
{
  "id": "182e2dc0-d877-41cc-8676-7203074f92fc"
}
```

PATCH

medical-appointment-backend-2xx0.onrender.com/api/appointments/182e2dc0-d877...

Send

Docs

Params

Auth

Headers (9)

Body

Scripts

Settings

Cookies

raw

JSON

Beautify

```
1 {
2   "id": "182e2dc0-d877-41cc-8676-7203074f92fc"
3 }
4
```

Body

200 OK • 677 ms • 799 B • 🌐 • ⋮

{ } JSON

Preview

Visualize

🔍

🔍

🔍

🔍

🔍

```
1 {
2   "id": "182e2dc0-d877-41cc-8676-7203074f92fc",
3   "patient_user_id": "c47e844c-1563-4314-9aaa-44d6c518a1b7",
4   "doctor_id": "8a6171c3-ba10-4f9b-8b4b-73f27c9169b6",
5   "scheduled_start": "2025-11-20T10:00:00+00:00",
6   "scheduled_end": "2025-11-20T15:30:00+00:00",
7   "status_id": 1,
8   "reason": "Chequeo",
9   "created_at": "2025-11-12T11:14:49.841325+00:00",
10  "updated_at": "2025-12-19T04:54:13.692404+00:00",

```

Find View

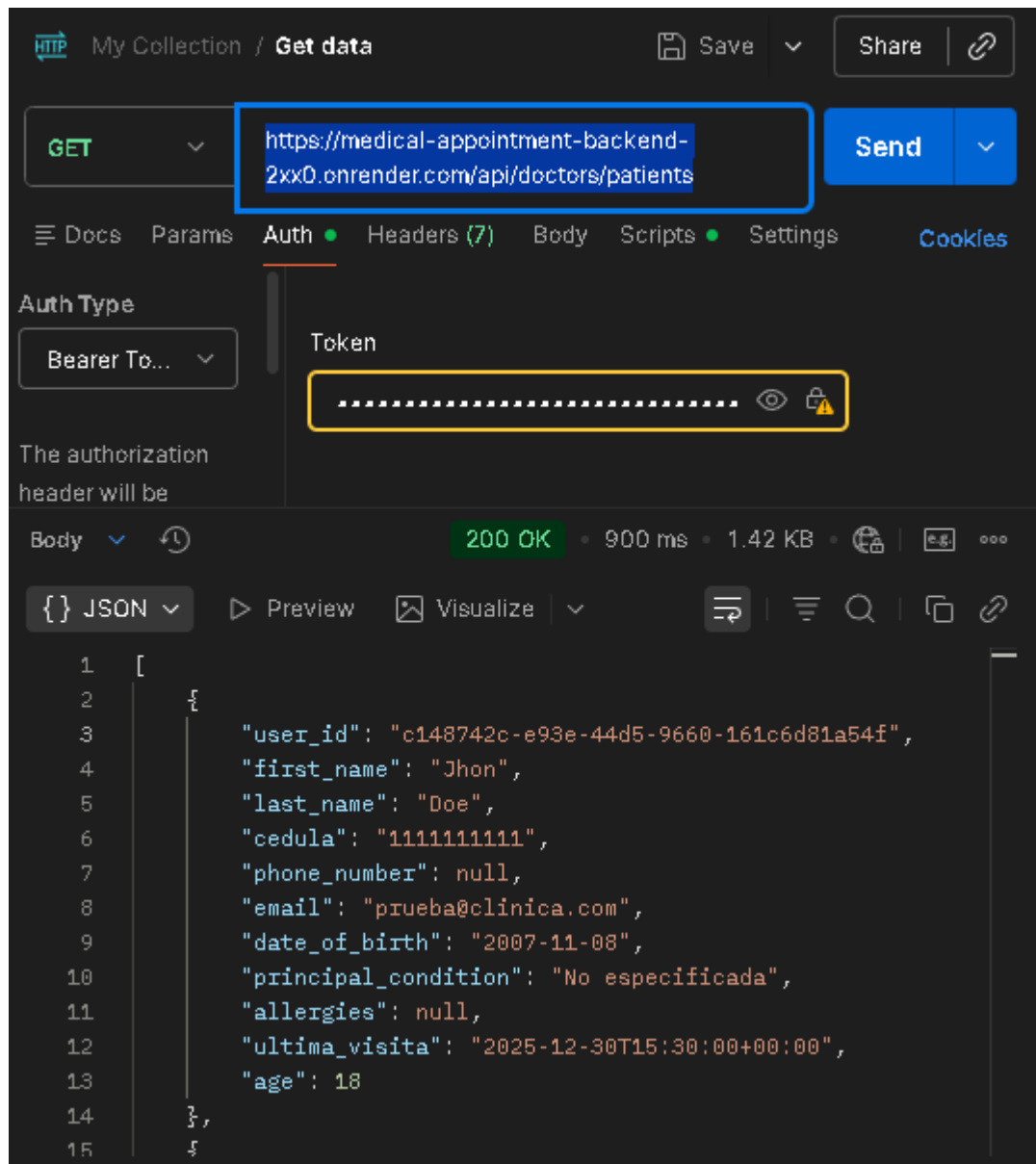
Find and replace

Console

Terminal

Appointment — GET /api/doctors/schedule (doctor)

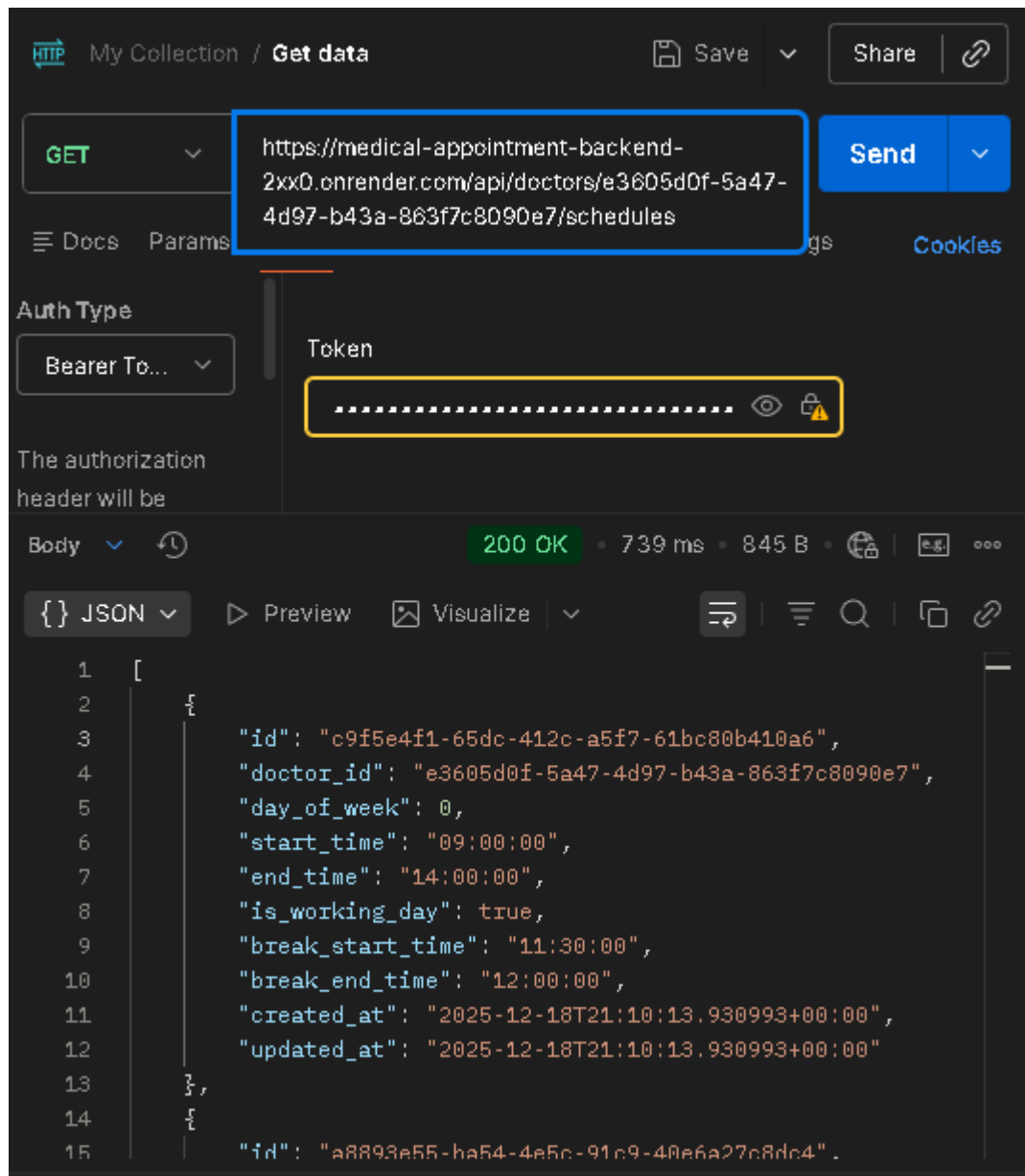
URI:<https://medical-appointment-backend-2xx0.onrender.com/api/doctors/schedule>



Schedule settings — GET /api/doctors/:id/schedules (Autenticado)

URI:

<https://medical-appointment-backend-2xx0.onrender.com/api/doctors/e3605d0f-5a47-4d97-b43a-863f7c8090e7/schedules>



Schedule settings — GET /api/appointments/doctor (doctor)

URI: <https://medical-appointment-backend-2xx0.onrender.com/api/appointments/doctor>

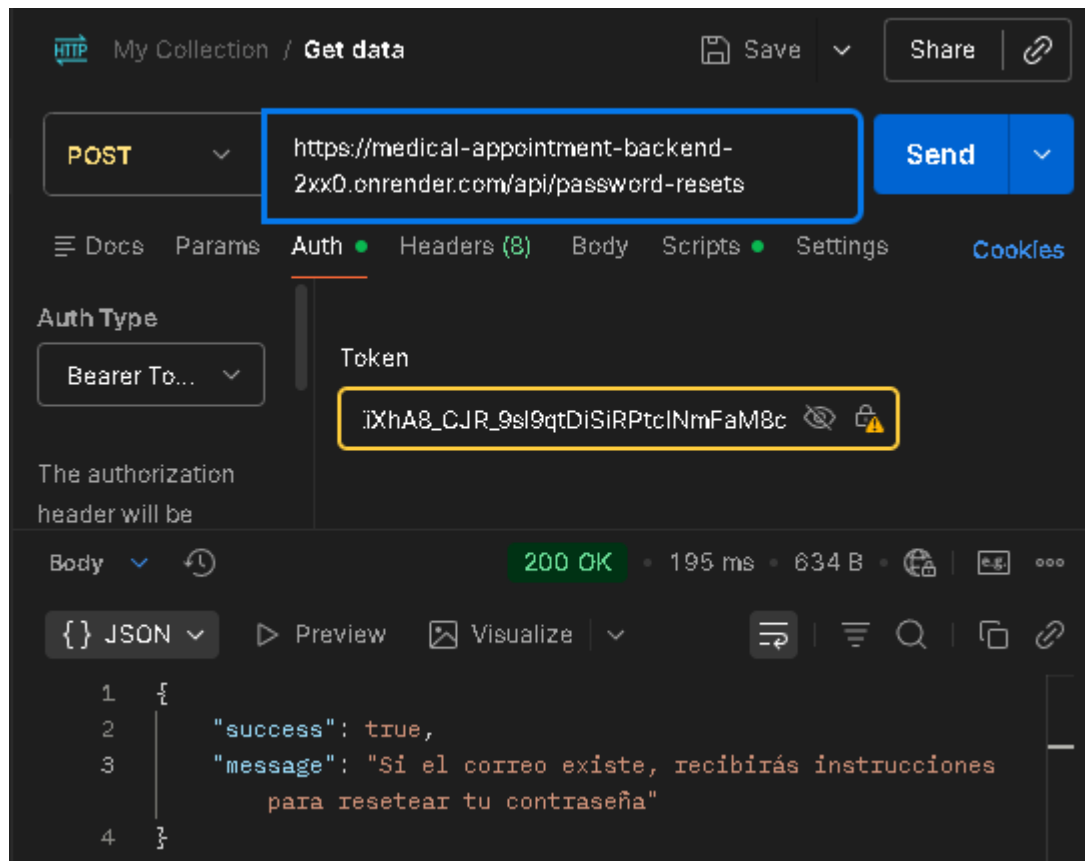
The screenshot shows a REST client interface with the following details:

- Method:** GET
- URL:** `https://medical-appointment-backend-2xx0.onrender.com/api/appointments/doctor`
- Response Status:** 200 OK
- Response Time:** 674 ms
- Response Size:** 832 B
- Response Body (JSON):**

```
[
  {
    "id": "761b9ac6-5932-4618-b258-4563ca6de09e",
    "patient_name": "Jhon Doe",
    "patient_email": "prueba@clinica.com",
    "scheduled_start": "2025-12-22T15:30:00+00:00",
    "scheduled_end": "2025-12-22T16:00:00+00:00",
    "reason": "Seguimiento",
    "status_id": 1,
    "status_label": "Programada",
    "status_code": "scheduled",
    "room_id": "e5e2a4e4-5e5e-4b4a-e5b5-7a8f1e3f6c5c",
    "room_name": "Consultorio Traumatología 1",
    "room_number": "T-301",
    "created_at": "2025-12-18T21:21:48.195926+00:00",
    "updated_at": "2025-12-18T21:21:48.195926+00:00"
  }
]
```

Email — `emailService.sendPasswordResetEmail()`

URI: <https://medical-appointment-backend-2xx0.onrender.com/api/password-resets>



Feature 3: Administrative Reporting and Billing

Req. ID	Grouped Requirement Description	Roles Involved	Type
RF-A. 3	Generation of Performance Reports and Comparisons (TC, AR, ACT, CR, PSI, OEI).	Administrator	Functional (Reporting/Analysis)
RF-A. 4	Dynamic Billing Generation based on Services, Duration, and Insurance.	Administrator	Functional (Billing)

RF-A.5	Transaction Audit Logging of all critical actions and timestamps.	Administrator	Functional (Compliance/Audit)
RNF-A.4	Compliance with Health Data Privacy Regulations (HIPAA/GDPR).	Administrator	Non-Functional (Legal)
RNF-C.2	Maintainability and Scalability of Source Code.	Development Team	Non-Functional (Maintainability)

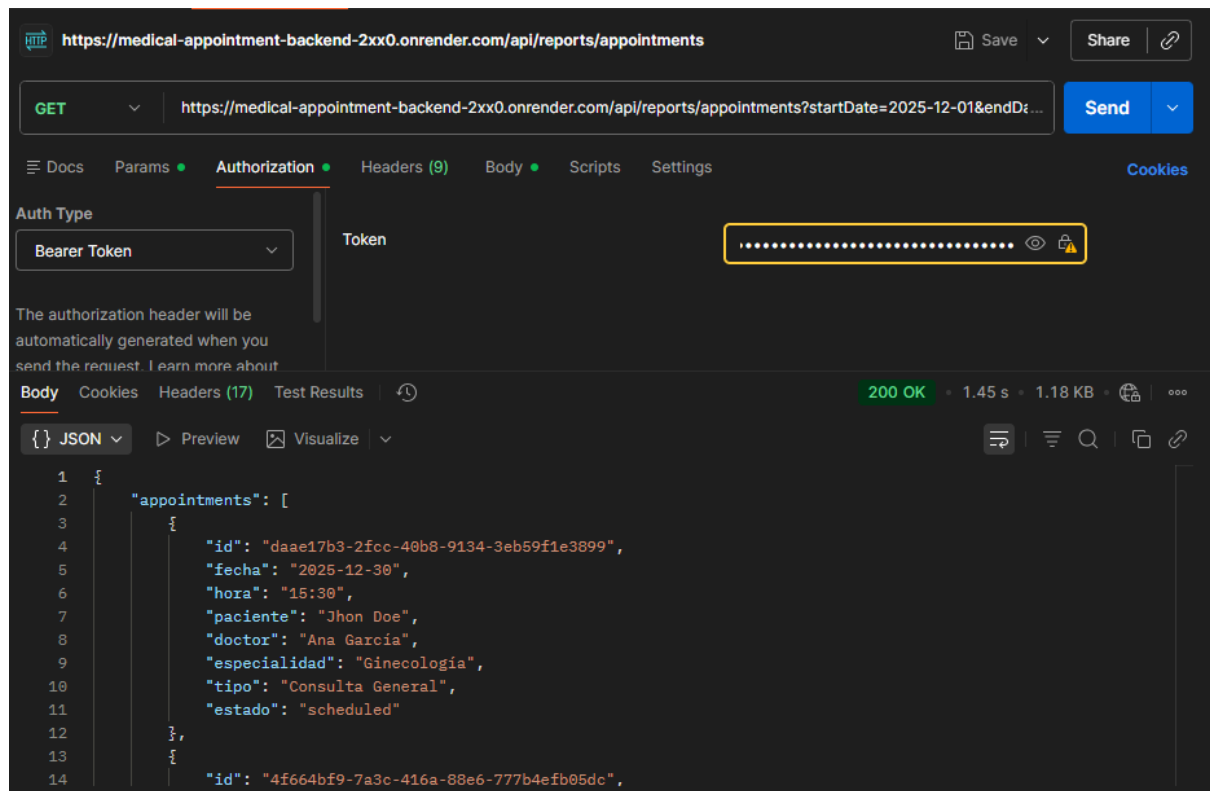
Get Appointments Report

Method: GET

Description: Retrieves detailed appointment information for a specific date range with comprehensive statistics including total consultations, unique patients, and consultation types breakdown.

Role Required: doctor or admin

medical-appointment-backend-2xx0.onrender.com/api/reports/appointments?startDate=2025-12-01&endDate=2025-12-31&period=mes



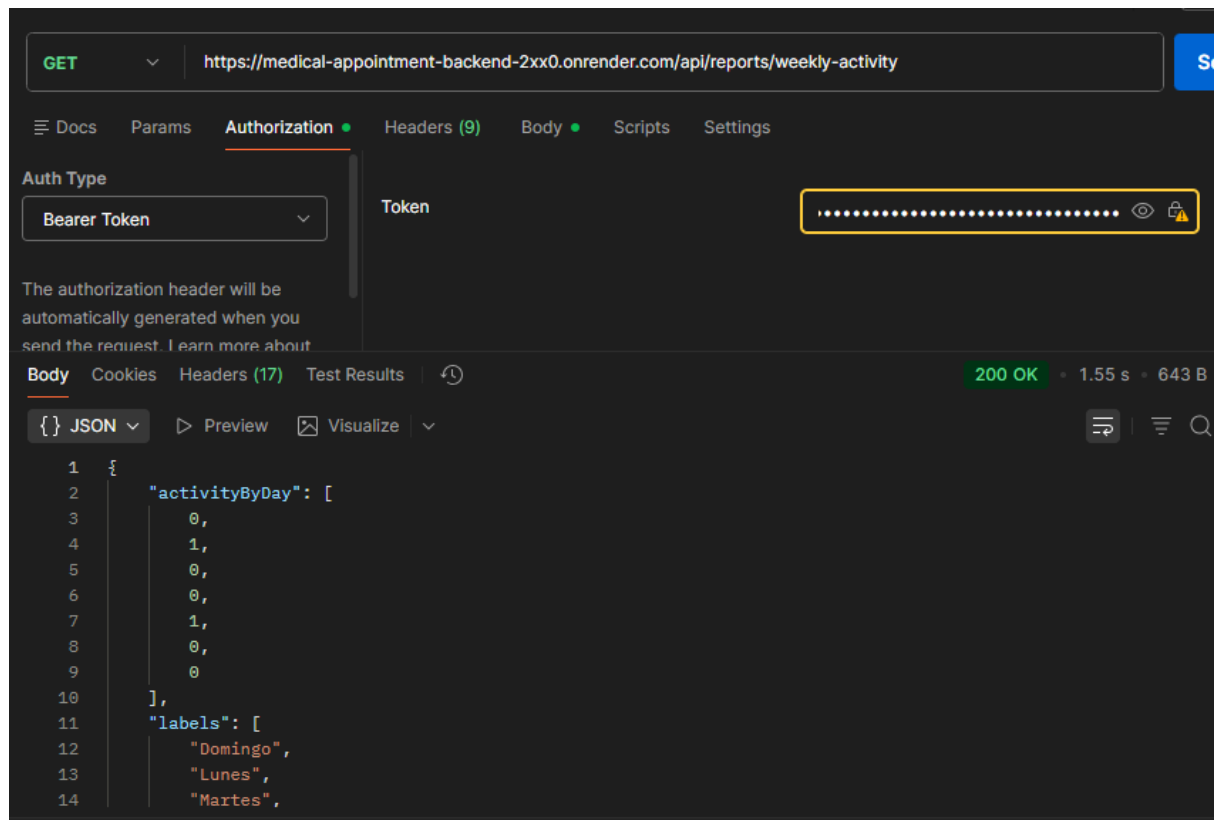
Get Weekly Activity Report

Method: GET

Description: Returns appointment activity grouped by day of the week for the current week, useful for visualizing workload distribution.

Role Required: doctor or admin

medical-appointment-backend-2xx0.onrender.com/api/reports/weekly-activity



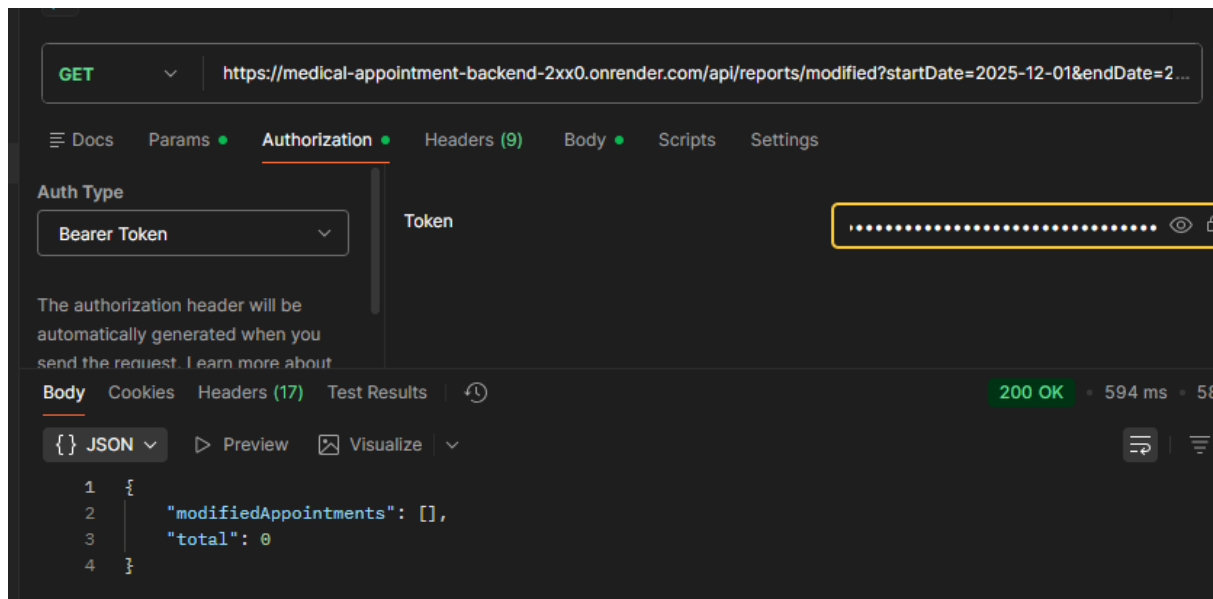
Get Modified Appointments Report

Method: GET

Description: Retrieves appointments that were cancelled or rescheduled within a date range, including reasons for modifications.

Role Required: doctor or admin

<https://medical-appointment-backend-2xx0.onrender.com/api/reports/modified?startDate=2025-12-01&endDate=2025-12-31&status=canceladas>



Create Billing

Method: POST

Description: Generates a new billing record with automatic calculation of total amount based on base amount, specialty multiplier, duration multiplier, insurance discount, and additional charges.

Role Required: admin

```
{  "appointment_id": "uuid-appointment",  "patient_user_id": "uuid-patient",  "doctor_id": "uuid-doctor",  "base_amount": 50.00,  "specialty_multiplier": 1.5,  "duration_multiplier": 1.2,  "insurance_discount_percentage": 20,  "additional_charges": 10.00,  "notes": "Consulta especializada de cardiología"}
```

<https://medical-appointment-backend-2xx0.onrender.com/api/billings>

```
{
  "appointment_id": "748d547a-1ca3-4c75-8e67-9d4306931286",
  "patient_user_id": "c47e844c-1563-4314-9aaa-44d6c518a1b7",
  "doctor_id": "4cba49e8-382e-4955-ba27-e781b2eba96a",
  "base_amount": 50.00,
  "specialty_multiplier": 1.5,
  "duration_multiplier": 1.2,
  "insurance_discount_percentage": 20,
  "additional_charges": 10.00,
  "notes": "Consulta especializada de cardiología"
}
```

The screenshot displays a REST client interface with the following details:

- URL:** `https://medical-appointment-backend-2xx0.onrender.com/api/billings`
- Method:** `POST`
- Body (raw):**

```
1 {
2   "appointment_id": "748d547a-1ca3-4c75-8e67-9d4306931286",
3   "patient_user_id": "c47e844c-1563-4314-9aaa-44d6c518a1b7",
4   "doctor_id": "4cba49e8-382e-4955-ba27-e781b2eba96a",
5   "base_amount": 50.00,
6   "specialty_multiplier": 1.5,
7   "duration_multiplier": 1.2,
8   "insurance_discount_percentage": 20,
```
- Response (JSON):**

```
1 {
2   "message": "Factura creada exitosamente",
3   "billing": {
4     "id": "7e4edfdb-0f64-4a21-ae1d-c1aad14fc5a5",
5     "appointment_id": "748d547a-1ca3-4c75-8e67-9d4306931286",
6     "patient_user_id": "c47e844c-1563-4314-9aaa-44d6c518a1b7",
7     "doctor_id": "4cba49e8-382e-4955-ba27-e781b2eba96a",
8     "base_amount": 50,
9     "specialty_multiplier": 1.5,
10    "duration_multiplier": 1.2,
11    "insurance_discount_percentage": 20.
```
- Status:** `201 Created`

Get Billing by ID

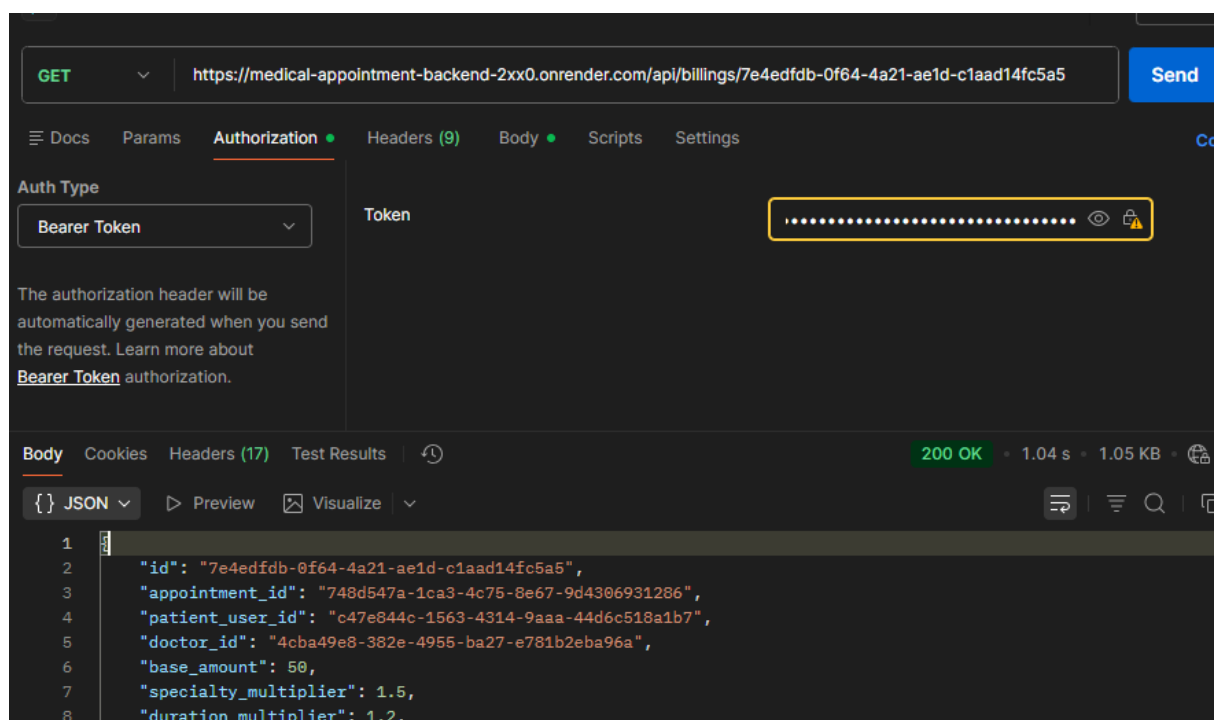
Method: GET

Description: Retrieves detailed billing information including appointment, patient, and doctor details.

Role Required: admin or doctor

<https://medical-appointment-backend-2xx0.onrender.com/api/billings/uuid-billing-here>

<https://medical-appointment-backend-2xx0.onrender.com/api/billings/7e4edfdb-0f64-4a21-ae1d-c1aad14fc5a5>



Get Billings with Filters

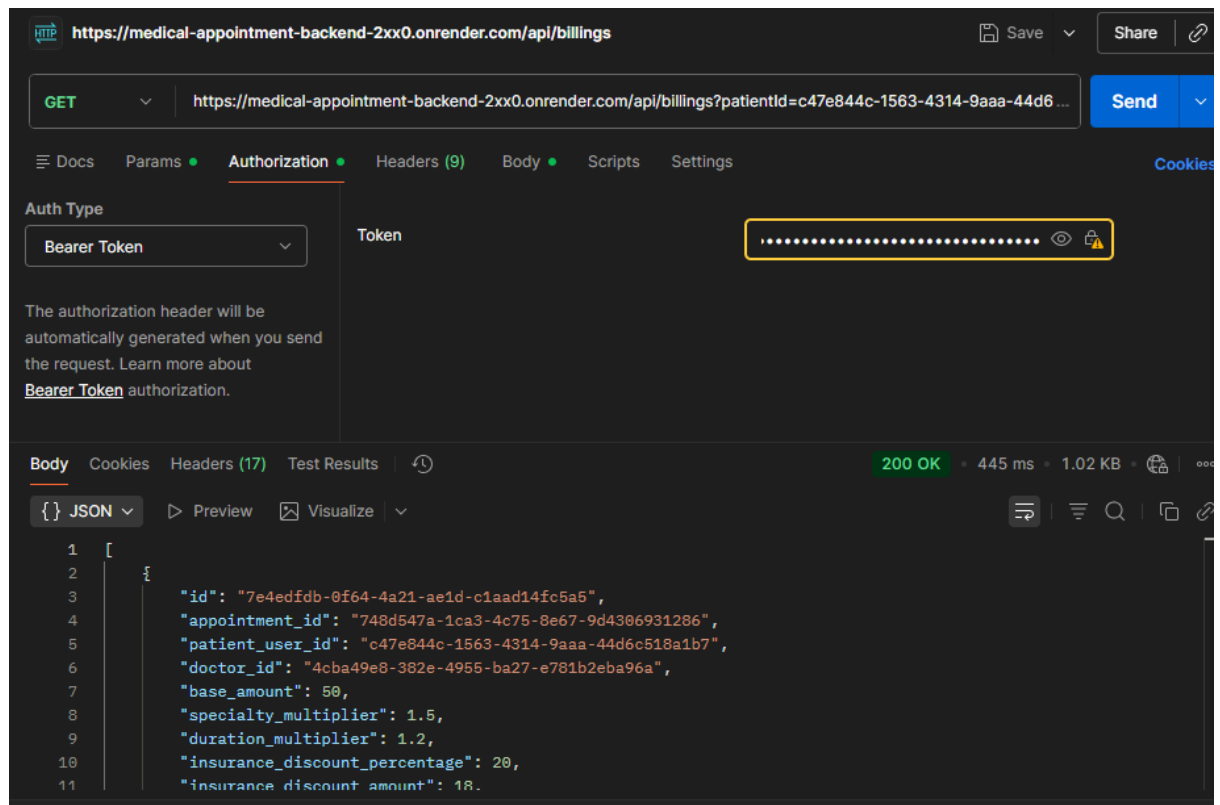
Method: GET

Description: Retrieves a list of billings with optional filters for patient, doctor, status, and date range.

Role Required: admin or doctor

`/api/billings?patientId={uuid}&doctorId={uuid}&status={status}&startDate={YYYY-MM-DD}&endDate={YYYY-MM-DD}`

<https://medical-appointment-backend-2xx0.onrender.com/api/billings?patientId=c47e844c-1563-4314-9aaa-44d6c518a1b7&status=pending>



Update Billing Status

Method: PUT

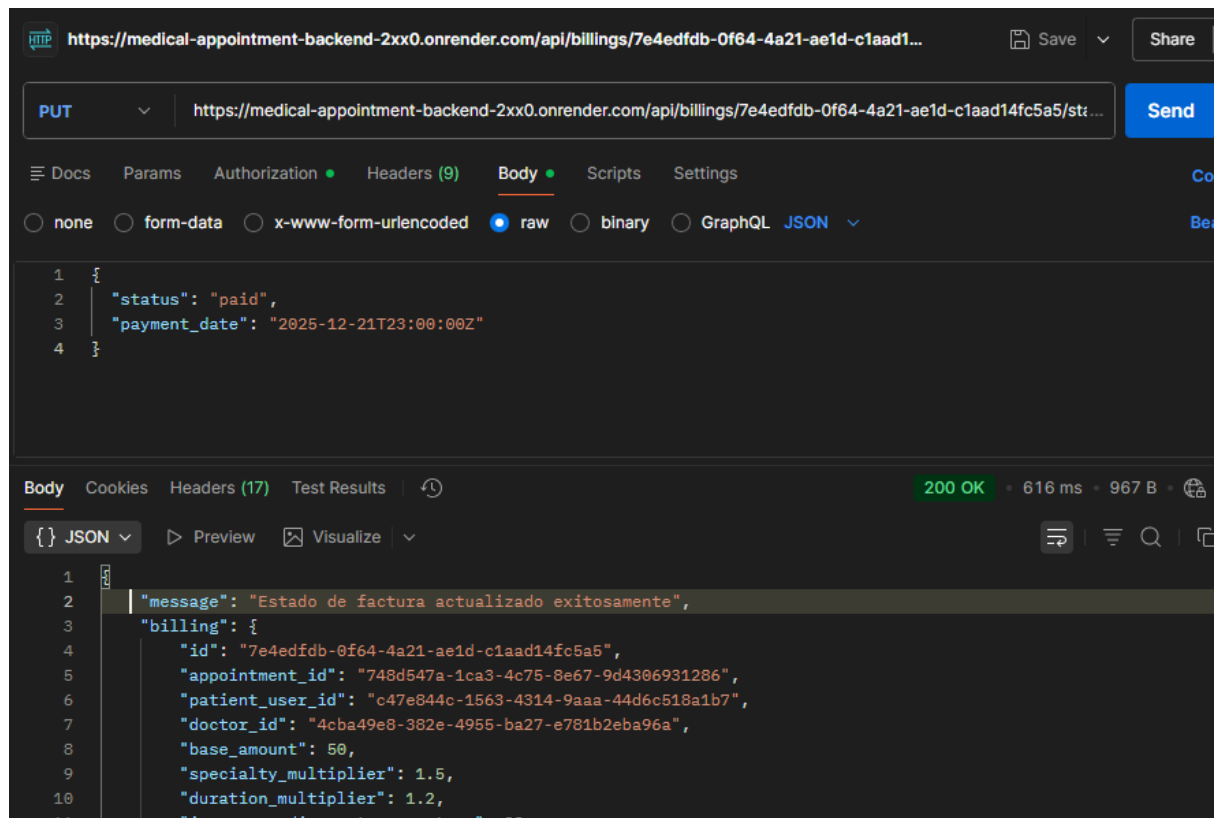
Description: Updates the payment status of a billing record, optionally recording the payment date.

Role Required: admin

<https://medical-appointment-backend-2xx0.onrender.com/api/billings/uuid-billing-here/status>

<https://medical-appointment-backend-2xx0.onrender.com/api/billings/7e4edfdb-0f64-4a21-ae1d-c1aad14fc5a5/status>

```
{
  "status": "paid",
  "payment_date": "2025-12-21T23:00:00Z"
}
```



Get Audit Logs with Filters

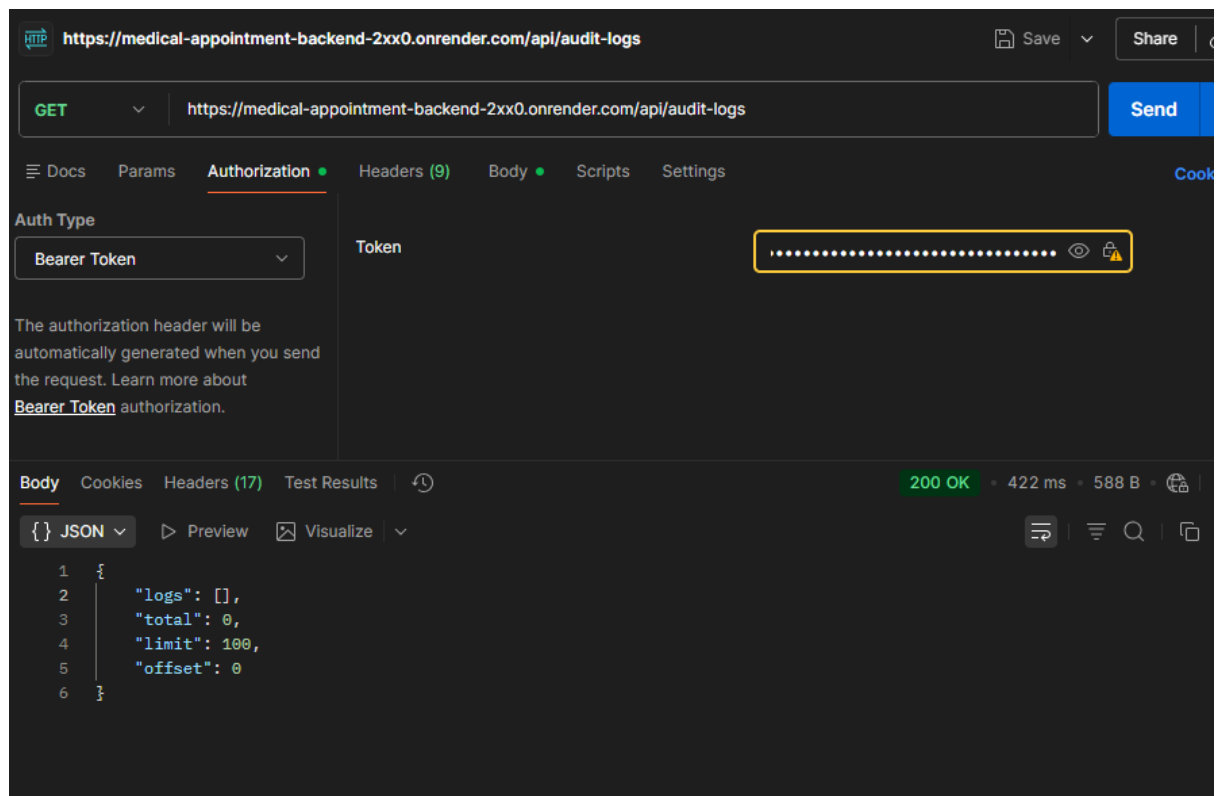
Method: GET

Description: Retrieves audit logs with comprehensive filtering options for compliance and security monitoring.

Role Required: admin

/api/audit-logs?userId={uuid}&action={action}&tableName={table}&startDate={YYYY-MM-DD}&endDate={YYYY-MM-DD}&limit={number}&offset={number}

<https://medical-appointment-backend-2xx0.onrender.com/api/audit-logs>



Create Audit Log (Manual)

Method: POST

Description: Manually creates an audit log entry for custom actions or external system integrations.

Role Required: admin

/api/audit-logs

```
{  "action": "manual_data_export",  "table_name": "appointments",  "record_id": "multiple",  "description": "Exportación manual de datos para auditoría externa",  "new_values": {    "exported_records": 500,
```

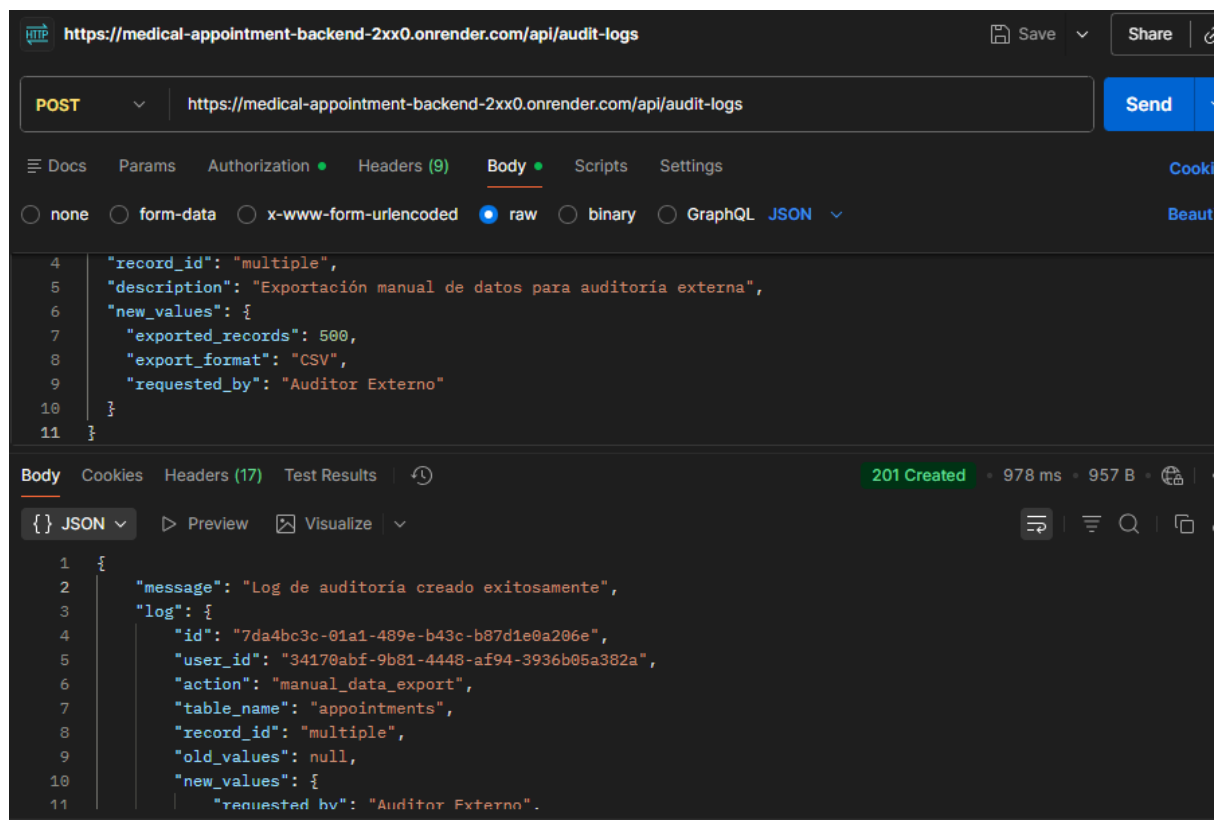
```
"export_format": "CSV",

"requested_by": "Auditor Externo"

}

}
```

<https://medical-appointment-backend-2xx0.onrender.com/api/audit-logs>



Get Audit Statistics

Method: GET

Description: Provides aggregated statistics of audit logs including top actions, most active users, and activity trends.

Role Required: admin

/api/audit-logs/stats?startDate={YYYY-MM-DD}&endDate={YYYY-MM-DD}

<https://medical-appointment-backend-2xx0.onrender.com/api/audit-logs/stats?startDate=2025-12-01&endDate=2025-12-31>

The screenshot shows a REST client interface with the following details:

- URL:** `https://medical-appointment-backend-2xx0.onrender.com/api/audit-logs/stats`
- Method:** GET
- Authorization:** Bearer Token. The token field is highlighted with a yellow box and contains a masked token (dots) and a warning icon.
- Response Status:** 200 OK, 653 ms, 690 B
- Response Body (JSON):**

```
1 {
2   "total": 1,
3   "topActions": [
4     {
5       "action": "manual_data_export",
6       "count": 1
7     }
8   ],
9   "topUsers": [
10    {
11      "userId": "34170ahf-9b81-4448-af94-3936b05a382a".
```